# Moving Apolar Constraints on Hodge Cycles of the Fermat Sextic Fourfold

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**Abstract**

We study the inverse problem of recovering algebraic cycles from moving period and apolar data on the Fermat sextic fourfold, whose rational Hodge conjecture is already known. Two disjoint Aoki–Shioda surfaces give a pencil of primitive classes with a uniform one-dimensional first-order Hodge excess. Exact second-order calculations separate the difference and sum classes: the chosen transverse difference direction is obstructed, whereas the sum direction extends through a finite formal Hodge jet of order six. The second-order transported apolar module has a unique quadratic generator transverse to the locus of split quadrics, together with four cubic generators modulo its multiples. The quadratic obstructs constructions requiring a split degree-two apolar generator but does not produce a cycle. Using the established complete-intersection period formula, we give a sufficient splitting criterion: a full apolar complete-intersection identity with the stated splitting and equation-membership hypotheses gives a $(2,3,3)$ surface representing the primitive functional. A fixed central union with the expected Hilbert polynomial has embedded Hilbert tangent dimension 116 and fails the incidence tangent equations for the chosen hypersurface direction. Standalone exact SageMath and Singular sources accompany these finite statements, with a current rerun of the second-order apolar calculation. The results identify which geometric input is missing between a finite Hodge jet and an actual cycle. They neither prove an all-orders extension of the jet nor give a new case of the Hodge conjecture.

## Moving Apolar Constraints on Hodge Cycles of the Fermat Sextic Fourfold

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### 1. Control problem

Let

$$X_6^4 = \{x_0^6 + \cdots + x_5^6 = 0\} \subset \mathbf{P}^5.$$

The rational Hodge conjecture for this Fermat sextic is known, and explicit linear, Aoki–Shioda and type-3 cycles span its primitive $(2,2)$-space [1]. We therefore use $X_6^4$ to test a harder inverse question:

> when a moving Hodge functional develops an apolar ideal with the shape of a complete intersection, does that signal an actual algebraic cycle?

The period-ideal framework follows [2], and the complete-intersection period formula used below is [3]. Movasati–Sertöz [2, Proposition 2.5 and Corollary 2.6], crediting Dan, already identify the period ideal of a complete intersection. Villaflor Loyola [3, Theorems 1.1–1.2] gives its explicit periods. The annihilator framework and the rank 122 of the $(2,3,3)$ case also occur in [4, Section 5]. Our contribution is the specific Aoki–Shioda pencil, its difference/sum separation, and the transported quadratic; Theorem 5.1 formulates a sufficient splitting condition for this calculation using those established period results.

Higher-order Hodge-locus obstructions already occur in [5], and [6, Section 4] explains why unrestricted period-annihilator inversion fails. The finite pencil and moving-apolar computations here concern a different explicit geometric family. The answer has two parts. A finite Hodge jet can expose a meaningful obstruction and can suggest a candidate geometry, but it does not by itself give a cycle. Conversely, a full apolar complete-intersection identity with the right splitting does give a cycle. The paper separates these two statements.

## 2. The disjoint Aoki–Shioda pencil

Let

$$K = \mathbf{Q}(\mu, a), \qquad \mu = \zeta_{12}, \qquad a^3 = 2,$$

and consider the two $(1, 2, 3)$ complete-intersection surfaces

$$\begin{aligned} S &= V(x_0^2 - a x_1 x_2,\ x_1^3 + x_2^3 + \mu^3 x_3^3,\ x_4 - \mu x_5), \\ S' &= V(x_1^2 - a x_0 x_2,\ x_0^3 + x_2^3 + \mu^3 x_4^3,\ x_5 - \mu x_3). \end{aligned} \tag{2.1}$$

They are related by a coordinate permutation and are disjoint. The normal-bundle calculation gives

$$c_2(N_{S/X_6^4}) = 11h^2|_S, \qquad S^2 = 66,$$

and the primitive intersection matrix is

$$\begin{pmatrix} 60 & -6 \\ -6 & 60 \end{pmatrix}. \tag{2.2}$$

In particular the two primitive classes are linearly independent.

The projective residue sign of the coordinate permutation is odd. Consequently, in a fixed residue convention, the raw pencil

$$P_t = P_0 + t P_1^{\mathrm{raw}} \tag{2.3}$$

represents the geometric class $[S] - t[S']$. Thus $t = 1$ is the difference class and $t = -1$ is the sum class. This sign is essential for interpreting the second-order calculation.

**Proposition 2.1 (uniform first-order excess)**

Let $M_t$ be the residue pairing matrix associated to $P_t$. Over the algebraic closure, for every $t \neq 0$,

$$\operatorname{rank} M_t = 122, \qquad \dim \ker M_t = 304,$$

whereas the common kernel of $M_0$ and $M_1^{\mathrm{raw}}$ has dimension 303. Thus every nonzero member of the pencil has exactly one extra first-order Hodge direction.

**Proof** The matrices act on the 426-dimensional degree-six monomial space. Exact sparse block decomposition shows that all quotient blocks have determinants equal to nonzero constants times powers of $t$; the unique exceptional 23-dimensional block has a nonzero principal minor proportional to

$$8(1 - \mu^2) a\, t^{10}.$$

This gives the lower rank bound for all $t \neq 0$. The explicit moving kernel

$$G_t = q(H_0 + tH_1), \qquad q = x_2^2 - a x_0 x_1,$$

with $H_0, H_1$ as in the reproducibility bundle, gives the matching upper bound. The dimensions follow. □

The excess is not a sufficient condition for a new geometric branch. It must be tested at the next order.

## 3. Second-order separation

Write a deformation as

$$F_s = F + sG + s^2 H + O(s^3).$$

The pole-order expansion is

$$\frac{q\Omega}{F_s^2} = \frac{q\Omega}{F^2} - 2s\frac{qG\Omega}{F^3} + s^2\left(3\frac{qG^2\Omega}{F^4} - 2\frac{qH\Omega}{F^3}\right) + O(s^3). \tag{3.1}$$

Using Griffiths reduction gives the exact second-order Hodge equations.

**Theorem 3.1 (difference/sum separation)**

For the pencil (2.3), the transverse first-order direction at the difference class $[S] - [S']$ has a nonzero second-order residue obstruction. The corresponding direction at the sum class $[S] + [S']$ solves the second-order equations and extends to the verified finite formal Hodge jet recorded through order six.

**Proof** In the coefficient and generator convention of the accompanying difference script, a nonzero second-order obstruction is

$$\frac{2}{3}(1-\mu^2)a \neq 0.$$

Thus the entire non-common first-order coset is blocked at second order. At the sum specialization, the same recurrence has a solution through order six; at each order the 426-dimensional right-hand side lies in the image of the first-order matrix, with augmented rank equal to 122. The calculation is over $K$ and uses exact residue reduction. □

The result is deliberately finite. It proves neither a convergent analytic branch nor a formal solution to all orders. In particular, a rank drop at first order is not a cycle construction.

## 4. The transported apolar ideal

For the sum direction, use the truncated base

$$A = K[s]/(s^3).$$

Let $L_s$ be the transported residue functional, including the Griffiths reductions of the deformed equation. Define

$$I_{\delta_s,r} = \{g \in S_r \otimes A \,:\, L_s(gq) = 0 \text{ for every } q \in S_{12-r} \otimes A\}. \tag{4.1}$$

The complete moving quotient, not a fixed central Jacobian quotient, is required in (4.1).

**Theorem 4.1 (irreducible quadratic and cubic generators)**

The degree-two moving apolar module is free of rank one and can be normalized as

$$\begin{aligned} q(s) = {} & (x_4 - \mu x_5)(x_3 - \mu^{-1}x_5) \\ & + \frac{1-\mu^2}{6}s(x_2^2 - a x_0 x_1) \pmod{s^3}. \end{aligned} \tag{4.2}$$

The first-order term of (4.2) is transverse to the tangent space of split quadratic forms through $q(0)$. Hence the moving quadratic cannot be factored into two moving linear forms, even after extending the coefficient field. Moreover,

$$I_{\delta_s,3} \simeq A^{10}, \qquad I_{\delta_s,3}/q(s)S_1 \simeq A^4. \tag{4.3}$$

#### Proof

The coefficient matrices of (4.1) have Toeplitz kernel dimensions $(1,1,1)$ in degrees two and $(10,10,10)$ in degree three. The degree-two tangent space to the factorization locus is

$$(x_4 - \mu x_5)S_1 + (x_3 - \mu^{-1}x_5)S_1.$$

It has rank 11, while adjoining the first-order term in (4.2) gives rank 12. Thus the first-order term is not tangent to the split locus. Equivalently, the two disjoint variable blocks of the generic quadratic have total rank five, so no higher-order lift can make it a product of linear forms.

The degree-three kernel calculation gives rank ten. Multiplication by the quadratic generator has rank six, and the quotient has rank four. All statements are exact over $A$. □

The apolar fingerprint is therefore one nonsplit quadratic plus four cubic minimal generators. In particular, it excludes any construction whose degree-two apolar generator is required to remain a product of two moving linear forms specializing to the two factors of $q(0)$. The assertion is exactly this factorization obstruction; no exclusion of other cycle constructions follows. A quadric–cubic flag would require further input.

## 5. A sufficient theorem that does produce a cycle

The finite calculations become algebraic only after replacing them by a full-order complete-intersection statement.

**Theorem 5.1 (moving-apolar-to-cycle specialization)**

Let $K$ be a characteristic-zero field,

$$S = K[x_0, \dots, x_5], \qquad X = V(F) \subset \mathbf{P}^5_K, \qquad \deg F = 6,$$

with $X$ smooth. Let $\zeta$ be a nonzero primitive Hodge class and let $L_\zeta : (S/J_F)_{12} \to K$ be its residue functional. Put $I = \mathrm{Ann}(L_\zeta)$. Assume:

1. $I$ is an Artin complete intersection with minimal generator degrees $(2, 3, 3, 3, 3, 4)$;
2. $J_F \subset I$;
3. $F \in (I^2)_6$;
4. in
$$(I^2/\mathfrak{m}I^2)_6 \simeq (E_2 \otimes E_4) \oplus \mathrm{Sym}^2(E_3), \qquad E_d = (I/\mathfrak{m}I)_d,$$
the class of $F$ has the form $\gamma qr + Q$, with $\gamma \neq 0$, and $Q$ nondegenerate of Witt index two on $E_3$.

Then there is a $(2, 3, 3)$ complete-intersection surface

$$Z = V(g_0, g_1, g_2) \subset X, \qquad \deg Z = 18,$$

and $\lambda \in K^\times$ such that

$$[Z]_{\mathrm{prim}} = \lambda\zeta. \tag{5.1}$$

If the data are defined over $\mathbf{Q}$ with a common residue normalization, then $\lambda \in \mathbf{Q}^\times$.

**Proof** Since the ideal is a complete intersection, the degree-six part of $I^2/\mathfrak{m}I^2$ has only the 2+4 and 3+3 contributions. The split hypothesis permits a choice of cubic generators $g_1, g_2, h_1, h_2$ with

$$Q = g_1h_1 + g_2h_2.$$

The difference between $F$ and $\gamma qr + g_1h_1 + g_2h_2$ lies in $(\mathfrak{m}I^2)_6$, hence can be absorbed into a new degree-four generator

$$h_0 = \gamma r + qA_2 + \sum_i \ell_i c_i.$$

Thus

$$F = qh_0 + g_1h_1 + g_2h_2. \tag{5.2}$$

The three equations $q, g_1, g_2$ define a $(2, 3, 3)$ complete intersection contained in $X$. Its residue functional has annihilator $I$, the same as $L_\zeta$. The socle degree is

$$2 + 3 + 3 + 3 + 3 + 4 - 6 = 12,$$

and the socle is one-dimensional. Hence the two nonzero residue functionals are scalar multiples, giving (5.1). □

This sufficient splitting formulation applies the established complete-intersection period formula [2, Proposition 2.5 and Corollary 2.6; 3, Theorems 1.1–1.2]. It specifies the additional full-order identity needed to pass from this moving calculation to a cycle. If the four hypotheses hold over the function field of an algebraic Hodge-locus curve, properness of the relative Hilbert scheme specializes the resulting surface to the original fibre.

## 6. Incidence obstruction for the central union

Put $\ell = x_4 - \mu x_5$, $\ell' = x_5 - \mu x_3$ and $T = V(\ell, \ell', F)$. Consider the central union

$$Y_0 = S \cup S' \cup T, \qquad I_{Y_0} = I_S \cap I_{S'} \cap (\ell, \ell', F).$$

Every component lies in $X_6^4$, so $F \in I_{Y_0}$. The supplied Singular calculation checks this union over $\mathbf{F}_{109}$, with $\mu = 8$ and $a = 57$; it makes no assertion about a separate saturated punctured family. The union has degree 18 and the Hilbert polynomial of a $(2,3,3)$ surface. Its coordinate-ring Hilbert series is

$$H_{R/I_{Y_0}}(t) = \frac{1 + 3t + 5t^2 + 5t^3 + 3t^4 + t^5}{(1-t)^3}, \qquad R = K[x_0, \dots, x_5].$$

Over $K = \mathbf{Q}(\zeta_{12}, 2^{1/3})$, the incidence calculation uses generators of degrees $(2,3,3,4,4,6)$. Parametrizing their images by polynomials in $R_{d_i}$ gives 847 unknowns and a syzygy system of rank 423, hence a 424-dimensional space of polynomial representatives. Images in $(I_{Y_0})_{d_i}$ represent the zero tangent vector. Their total dimension, computed from the displayed Hilbert series, is

$$\sum_i \dim_K (I_{Y_0})_{d_i} = 1 + 8 + 8 + 33 + 33 + 225 = 308.$$

Consequently the embedded Hilbert tangent space has dimension

$$\dim_K \operatorname{Hom}_R(I_{Y_0}, R/I_{Y_0})_0 = 424 - 308 = 116.$$

Imposing the first-order hypersurface-incidence equation for the chosen transverse Hodge direction gives coefficient rank 537 and augmented rank 538. This system is inconsistent even before quotienting out zero representatives, so the quotient cannot supply a solution. Thus this particular central union cannot deform inside the chosen first-order hypersurface deformation. This is a central incidence obstruction, not a non-membership claim for another special surface and not an obstruction to every possible center.

## 7. Mathematical boundary

The results establish four distinct evidence levels:

| input | valid conclusion |
|---|---|
| first-order rank drop | candidate Hodge direction |
| finite Hodge jet | finite formal compatibility |
| full apolar CI hypotheses | actual $(2,3,3)$ algebraic surface |
| fixed central incidence rank 537/538 | that central union cannot realize the chosen direction |

In particular, the sextic control geometry reveals a general methodological point: apolar data can be a faithful cycle detector only after integrability, complete-intersection structure and the equation-membership identity have been proved. Neither a finite jet nor an Artin Hilbert function alone supplies those conditions.

## Reproducibility

The supplement contains standalone SageMath and Singular source programs for the first-order pencil, difference obstruction, the sum arc through order six, second-order moving-apolar modules, the fixed central Hilbert-tangent incidence calculation, and the central union over $\mathbf{F}_{109}$. The second-order moving-apolar calculation was rerun on October 22, 2026. A quotient-only check on October 23 recomputed the six ideal-piece dimensions from the actual Gröbner basis over both $\mathbf{F}_{109}$ and $K$, confirming their sum 308. The large syzygy and incidence systems were not rerun; their outputs remain identified as historical recorded results. No wrapper requiring a private project or unavailable temporary data is used.

Run the SageMath scripts from the extracted archive root, with SageMath 10.9 or compatible. For the central incidence calculation use `Y0_EXACT=1 sage calculations/bundles/aoki-sextic-y0-hilbert-tangent/aoki-y0-hilbert-tangent.` omitting `Y0_EXACT` runs its finite-field control. Run the central-union Hilbert-series check with `Singular -q calculations/bundles/aoki-sextic-ci-special-fiber/aoki-sextic-ci-special-fiber.sing`. Full commands and the scope of each calculation appear in the supplement README.

These finite computations do not verify the full-order complete-intersection hypotheses in Theorem 5.1 and do not establish convergence of the formal Hodge jet. Copyright remains with the author, subject to rStation's publication and review terms.

## References


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